FAST
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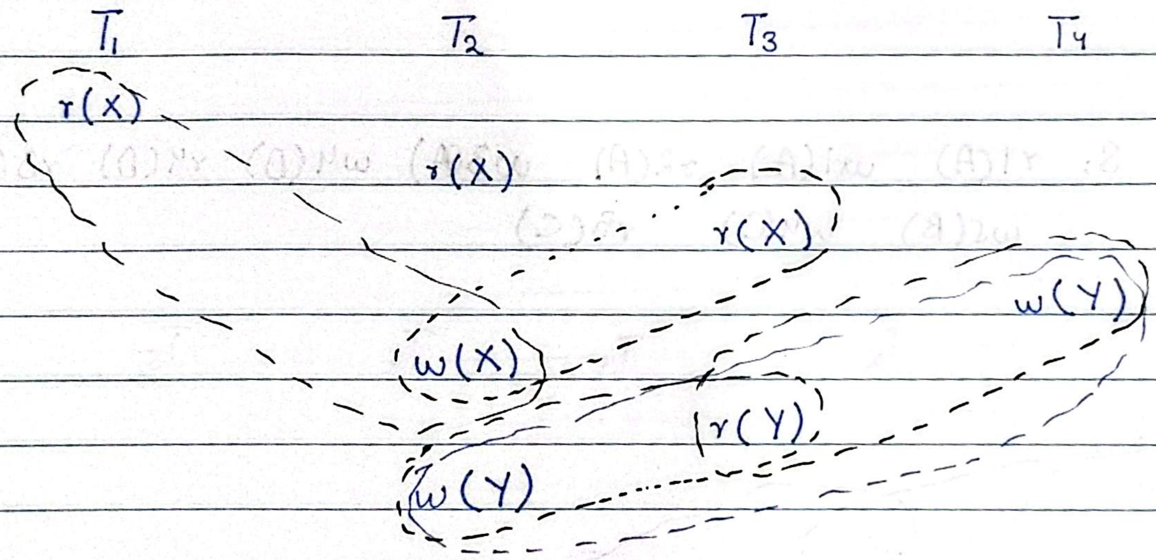
Transactions Consider the following schedule of four transactions T_1, T_2, T_3, T_4

S: r1(X), r2(X), r3(X), w4(Y), w2(X), r3(Y), w2(Y)

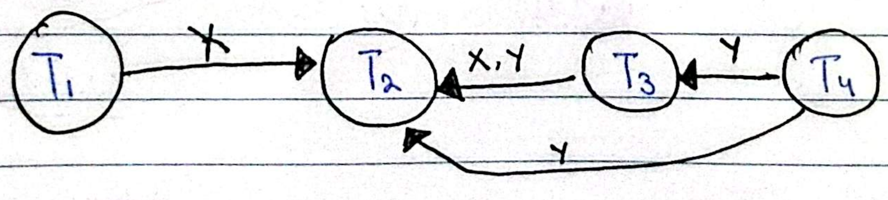
Draw the serializability (precedence graph) for this schedule. State whether this has conflict or not. If the schedule is serializable, write down the equivalent serial schedule, otherwise explain why it is not.

Sol:- Step1:- Write everything in Transaction form

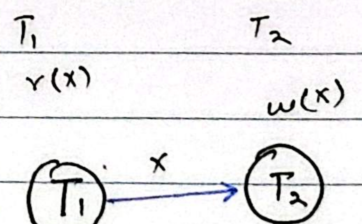


★ For same variable only r() r() is correct everything else is false. r() w() X w() r() X w() w() X

Step 2: Draw precedence graph



How to connect nodes



Tail at conflict starter
Head at conflict ender

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Step 3:- Write sequence :- If there is a cycle in the graph there will be no serializable equivalent sequence

Otherwise you can serialize it.

* ★ Node having no incoming edge can start the sequence
* ★ Node having no outgoing edge can end the sequence.

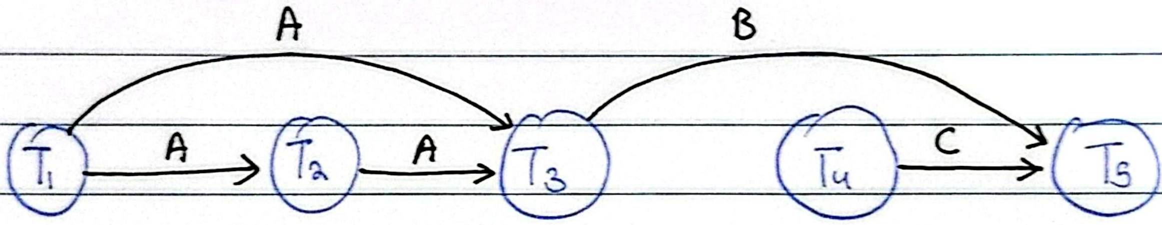
$T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_2$ there can be multiple answers

$T_4 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2$ etc

Q:- S: r1(A) w1(A) r2(A) w3(A) w4(D) r4(D) r3(B) w5(B) w4(C) r5(C)

Sol:-

T_1	T_2	T_3	T_4	T_5
r(A)				
w(A)				
	r(A)			
		w(A)		
			w(D)	
			r(D)	
		r(B)		
				w(B)
			w(C)	
				r(C)



As no cycle so it is serializable.

$\boxed{T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4 \rightarrow T_5}$